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References:

1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F Korth
2. Fundamentals of Database Systems by Elmsari, Navathe, 5th Edition Pearson

Faculty HOD Dean Academics 16/3/24

UNIT-3(Introductions to Schema Refinement)
LECTURE NO: -21

Database schemas tell the database what your data is and how to work with it. Database schemas help the database engine understand these patterns which allows it to enforce constraints on the data, respond with the right information when queried, and manipulate it in ways that users request.

Good schemas tend to reduce implicit information in favour of making it visible to the system and its users. Schemas in relational databases can reduce information redundancy,ensure data consistency, and provide the scaffolding and structures needed to access and join related data.

Definingphysicalvslogicalschemas

When talking about designing database schemas, **a logical schema** is a general design for organizing data into different categories, defining properties of the data, and determining the best structure for database items.

a **physical schema** is recognized as being the next step in the design process where implementation-specific details are worked out.

Schema refinement is just a fancy term for saying polishing tables. It is the last step before considering physical design with typical workloads:

1) Requirementanalysis:userneeds

2) Conceptualdesign:high-leveldescription,oftenusingE/Rdiagrams

3) Logicaldesign:fromgraphstotables(relational schema)

4) Schemarefinement:checkingtablesforredundanciesandanomalies

Let's see an example of redundancies and anomalies. Consider the following table where the client's name is the primary key.

employeeID	FirstName	LastName	Hire Date	Client
1	Mary	Marley	1/1/2005	James H.
2	Robert	Crosby	2/4/2008	Clark L.
3	Robert	Crosby	2/4/2008	Tina M.

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Introductions to Schema Refinement

Thetable ispresenting informationonemployees(salesreps)andtheirclients. If

we want to **insert data**, we notice that:

- eachrow requiresanentryintheclientfield.
- we can't insert data for newly hired sales reps until they've been assigned to one or more clients.
- if sales reps are in a training process, even if they've been already hired, they can't actually join the database because they need to have a delegated client… unless "dummy" clients are created.

Ifwewantto**updatedata**,wenoticethat:

- thesalesrepsnameisrepeatedforeachclient.
- what if, for a given client, we misspelled the name of the sales reps Crosby instead of Cosby… how can we edit that without affecting all the sales reps called Crosby?

If we want to **delete data,** what if Mary doesn't have a client anymore because she's taking a year off? We are forced to either

- createadummy client
- incorrectlyshowingherwitha clientshenolongerhandled
- deleteMary'srecord(even ifhowever,she'sstillanemployee)
- noticewecannothave"null"asaclientsince primaryfieldkeyscannotstorenull.

When we have to treat with schema refinement, we often notice that the main problem is redundancy. In order to identify schemas with such problems, we'll introduce the notion of functional dependencies: a relationship that exists when one attribute uniquely determines another attribute. A functional dependency is simply a new type of constraint between two attributes.

SaythatRisa relationwithattributesXand Y,we saythatthereisafunctional dependencyX -> Ywhen YisfunctionallydependentonX (whereX isthedeterminantsetandYisthe dependent attribute).

References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, PearsonEducation (2008).

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UNIT-3(FunctionalDependencies)
LECTURE NO: -22

WhatisFunctionalDependency?

Functional Dependency (FD) is a constraint that determines the relation of one attribute to another attribute in a Database Management System (DBMS). Functional Dependency helps to maintain the quality of data in the database. It plays a vital role to find the difference between good and bad database design.

A functional dependency is denoted by an arrow "→". The functional dependency of X on Y is represented by X → Y. Let's understand Functional Dependency in DBMS with example.

Example:

Employeenumber	EmployeeName	Salary	City
1	Dana	50000	SanFrancisco
2	Francis	38000	London
3	Andrew	25000	Tokyo

In this example, if we know the value of Employee number, we can obtain Employee Name,city, salary, etc. By this, we can say that the city, Employee Name, and salary are functionally depended on Employee number.

KeyTerms	Description
Axiom	Axiomsisasetofinferencerulesusedtoinferallthe functional dependencies on a relational database.
Decomposition	It is a rule that suggests if you have a table that appears to contain two entitieswhicharedeterminedbythesameprimarykeythenyoushould consider breaking them up into two different tables.
Dependent	Itisdisplayedontherightsideofthe functionaldependencydiagram.
Determinant	Itisdisplayedonthe leftside ofthe functionaldependencyDiagram.
Union	Itsuggeststhatiftwotablesareseparate,andthe PKisthesame,you should consider putting them. together

RulesofFunctionalDependencies

Below are the Three most important rules for Functional Dependency in Database:

Reflexiverule–.IfX isasetofattributesandYis_subset_ofX,thenX holdsavalue ofY.

Augmentationrule:Whenx-> yholds, andcisattribute set, thenac->bc alsoholds. Thatis adding attributes which do not change the basic dependencies.

Transitivity rule: This rule is very much similartothe transitive rule inalgebra if x-> y holds and y -> z holds, then x -> z also holds. X -> y is called as functionally that determines y.

TypesofFunctionalDependenciesinDBMS

TherearemainlyfourtypesofFunctionalDependencyinDBMS.Followingarethetypesof Functional Dependencies in DBMS:

- Trivialfunctionaldependency
- Non-Trivialfunctionaldependency
- Multivaluedfunctionaldependency
- Transitivefunctionaldependency

1. TrivialFunctionalDependency

InTrivialFunctionalDependency,adependentisalwaysasubsetofthe determinant.

i.e.IfX→Y andYisthesubsetofX, then itiscalledtrivialfunctional dependency.

Forexample,

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here,{roll_no,name}→nameisa trivial functionaldependency,sincethe dependentname is a subset of determinant set {roll_no, name}

Similarly,roll_no→roll_noisalsoanexampleoftrivialfunctionaldependency.

2. Non-trivialFunctionalDependency

InNon-trivialfunctionaldependency,thedependentisstrictlynotasubsetofthe determinant.

i.e.IfX→Y andYisnotasubsetof X,thenitiscalledNon-trivial functionaldependency.

Forexample,

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here, roll_no → name is a non-trivial functional dependency, since the dependent name is nota subset of determinant roll_no. Similarly, {roll_no, name} → age is also a non-trivial functional dependency, since age is not a subset of {roll_no, name}

3. MultivaluedFunctionalDependency

Multivalued dependency occurs in the situation where there are multiple independent multivalued attributes in a single table. A multivalued dependency is a complete constraint between two sets of attributes in a relation. It requires that certain tuples be present in a relation.

In Multivalued functional dependency, entities of the dependent set are not dependent oneach other.

i.e. If a → {b, c} and there exists no functional dependency between b and c, then it is called a multivalued functional dependency.

Forexample,

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here,roll_no→{name,age}isamultivaluedfunctional dependency,

4. TransitiveFunctionalDependency

Intransitivefunctionaldependency,dependentis indirectlydependentondeterminant.

i.e.Ifa → b & b → c,then accordingtoaxiom of transitivity,a → c. Thisis atransitive functional dependency.

For example,

enrol_no	name	dept	building_no
42	abc	CO	4
43	pqr	EC	2
44	xyz	IT	1
45	abc	EC	2

Here,enrol_no→deptanddept→ building_no,

Hence, according to the axiom of transitivity, enrol_no → building_no is a valid functional dependency. This is an indirect functional dependency, hence called Transitive functional dependency.

References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. FundamentalsofDatabaseSystems"byElmsari,Navathe,5thEdition,PearsonEducation (2008).

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UNIT-3(NormalFormsandFirst,SecondformofNormalization)
LECTURE NO: -23

WhatisNormalization?

Normalization is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies. Normalization rules divides largertables intosmallertables andlinks themusing relationships. Thepurpose of Normalisation in SQL is to eliminate redundant (repetitive) data and ensure data is stored logically.

Normalizationisusedformainlytwopurposes,

- Eliminatingredundant(useless)data.
- Ensuringdatadependenciesmakesensei.edataislogicallystored.

ProblemsWithoutNormalization

If a table is not properly normalized and have data redundancy then it will not only eat up extra memory space but will also make it difficult to handle and update the database, without facing data loss. Insertion, Updation and Deletion Anomalies are very frequent if database is not normalized. To understand these anomalies let us take an example of a Student table.

rollno	name	branch	hod	office_tel
401	Akon	CSE	Mr.X	53337
402	Bkon	CSE	Mr.X	53337
403	Ckon	CSE	Mr.X	53337
404	Dkon	CSE	Mr.X	53337

In the table above, we have data of 4 Computer Sci. students. As we can see, data for the fields branch, hod(Head of Department) and office_tel is repeated for the students who are in the same branch in the college, this is Data Redundancy.

There are three types of anomalies that occur when the database is not normalized. These are – Insertion, update and deletion anomaly. Let's take an example to understand this.

Example: Suppose a manufacturing company stores the employee details in a table named employee that has four attributes: emp_id for storing employee's id, emp_name for storing employee's name, emp_address for storing employee's address and emp_dept for storing the department details in which the employee works. At some point of time the table looks like this:

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

The above table is not normalized. We will see the problems that we face when a table is not normalized.

Update anomaly: In the above table we have two rows for employee Rick as he belongs to two departments of the company. If we want to update the address of Rick then we have to update the same in two rows or the data will become inconsistent. If somehow, the correct address gets updated in one department but not in other then as per the database, Rick wouldbehavingtwodifferentaddresses,whichis notcorrectandwouldleadtoinconsistent data.

Insert anomaly: Suppose a new employee joins the company, who is under training and currently not assigned to any department then we would not be able to insert the data into the table if emp_dept field doesn't allow nulls.

Delete anomaly: Suppose, if at a point of time the company closes the department D890 thendeleting the rows that are having emp_dept as D890wouldalsodelete the information of employee Maggie since she is assigned only to this department.

NormalForm	Description
1NF	Arelationisin1NF ifitcontainsanatomicvalue.
2NF	Arelationwillbein2NFifitisin1NFandallnon-keyattributesarefully functional dependent on the primary key.
3NF	Arelationwillbein3NFifitisin2NFandnotransitiondependencyexists.
4NF	A relation will be in 4NF if it is in Boyce Codd normal form and has no multi-valued dependency.
5NF	A relation is in 5NF if it is in 4NF and not contains any join dependency andjoining should be lossless.

ForatabletobeintheFirstNormalForm, itshouldfollowthefollowing4 rules:

- Itshouldonlyhavesingle(atomic)valuedattributes/columns.
- Valuesstored inacolumnshouldbeofthesamedomain
- Allthecolumnsinatable shouldhaveuniquenames.

emp_id	emp_name	emp_address	emp_mobile
101	Herschel	NewDelhi	8912312390
102	Jon	Kanpur	8812121212,9900012222
103	Ron	Chennai	7778881212
104	Lester	Bangalore	9990000123,8123450987

Two employees (Jon & Lester) are having two mobile numbers so the company stored them in the same field as you can see in the table above.

This table is not in 1NF as the rule says “each attributeof a table must have atomic (single) values”, the emp_mobile values for employees Jon & Lester violates that rule.

Tomakethetablecomplieswith1NFweshouldhavethedatalikethis:

emp_id	emp_name	emp_address	emp_mobile

101	Herschel	NewDelhi	8912312390
102	Jon	Kanpur	8812121212
102	Jon	Kanpur	9900012222
103	Ron	Chennai	7778881212
104	Lester	Bangalore	9990000123
104	Lester	Bangalore	8123450987

Secondnormalform(2NF)

Atableissaidtobe in2NF ifboththe followingconditions hold:

- Table isin1NF(Firstnormalform)

Inthesecondnormalform,allnon-keyattributesarefullyfunctionaldependentonthe primary key.

KeyAttribute:-

of Normalization

Uniquely identify a record or an instance on an entity. Required to retrieve a particular record from the entity table. non-prime (non-key) attributes are the attributes other thanthe primary key attributes.They can store a value any manytimes.

STUDENT(RegNo,Name,Address)

RegNoisprimarykey(KeyorprimeAttribute) Name

and address are Non-Key attribute

Let'sassume,aschoolcanstorethedataofteachersandthesubjectstheyteach.Ina school,

ateachercanteachmorethanonesubject.

TEACHER_ID	SUBJECT	TEACHER_AGE
25	Chemistry	30
25	Biology	30
47	English	35
83	Math	38
83	Computer	38

Inthegiven table,non-primeattributeTEACHER_AGEisdependent on TEACHER_IDwhich is a proper subset of a candidate key. That's why it violates the rule for 2NF.

Toconvertthegiventable into2NF,wedecomposeitintotwotables:

TEACHER_ID	TEACHER_AGE
25	30
47	35
83	38

TEACHER_ID	TEACHER_AGE
25	30
47	35
83	38

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References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. FundamentalsofDatabaseSystems"byElmsari,Navathe,5thEdition,Pearson Education (2008).

UNIT-3(Boyce-CoddNormalForms,ThirdNormalForm)
LECTURE NO: -24

A transitive dependency in a database is an indirect relationship between values in the same table that causes a functional dependency. To achieve the normalization standardof Third NormalForm (3NF),you must eliminate any transitive dependency.

{Book}->{Author}(ifweknowthebook,we knowtheauthorname)

{Author}-> {Author_age}Therefore,aspertheruleoftransitivedependency:

{Book}->{Author_age}shouldhold,

Book	Author	Author_age
Gameof Thrones	GeorgeR.R.Martin	66
HarryPotter	J.K.Rowling	49
DyingoftheLight	GeorgeR.R.Martin	66

ThirdNormalForm

Arelationwillbein3NFifitisin2NFandnotcontainanytransitivepartial dependency.

3NFisusedtoreducethedataduplication.Itisalsousedtoachievethedata integrity.

Ifthereisnotransitivedependencyfornon-primeattributes,thentherelation must be in third normal form.

A relationis inthird normalform ifit holds at least one ofthe following conditions for every non-trivial function dependency X → Y.

Xisasuper key.

Yisaprime attribute,i.e.,eachelementofYispartof somecandidatekey.

SuperKey:Thesetof attributeswhich can uniquelyidentifyatupleisknown as Super Key.

ForExample,STUD_NO,(STUD_NO,STUD_NAME)etc.

- Addingzeroormoreattributestocandidatekeygeneratessuperkey.
- Acandidatekeyisasuperkeybutviceversaisnottrue.

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Candidate Key:The minimalset of attributes whichcanuniquely identify a tuple is known as candidate key.

ForExample,STUD_NOinSTUDENTrelation.

- ThevalueofCandidateKeyisuniqueandnon-nullforevery tuple.
- Therecan bemorethanonecandidatekeyinarelation.

ExampleofSuper&CandidateKey

STUDENT

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY	STUD_AGE
1	RAM	9716271721	Haryana	India	20
2	RAM	9898291281	Punjab	India	19
3	SUJIT	7898291981	Rajsthan	India	18
4	SURESH		Punjab	India	21

Table 1

STUDENT_COURSE

STUD_NO	COURSE_NO	COURSE_NAME
1	C1	DBMS
2	C2	Computer Networks
1	C2	Computer Networks

Table 2

Example:-EMPLOYEE_DETAIL

EMP_ID	EMP_NAME	EMP_ZIP	EMP_STATE	EMP_CITY
222	Harry	201010	UP	Noida

333	**Stephan**	**02228**	**US**	**Boston**
444	**Lan**	**60007**	**US**	**Chicago**
555	**Katharine**	**06389**	**UK**	**Norwich**
666	**John**	**462007**	**MP**	**Bhopal**

Superkeyinthetableabove:

{EMP_ID},{EMP_ID,EMP_NAME},{EMP_ID,EMP_NAME, EMP_ZIP}.....soon.

Candidatekey:{EMP_ID}

Non-primeattributes:Inthegiventable,allattributesexceptEMP_IDarenon- prime.

Here, EMP_STATE &EMP_CITY dependent onEMP_ZIP and EMP_ZIP dependent on EMP_ID.

Thenon-primeattributes(EMP_STATE,EMP_CITY)transitivelydependentonsuper key (EMP_ID). It violates the rule of third normal form.

That'swhyweneedtomovetheEMP_CITYandEMP_STATEtothenew <EMPLOYEE_ZIP>table,withEMP_ZIPasaPrimarykey.

EMPLOYEE Table: -

EMP_ID	EMP_NAME	EMP_ZIP
222	**Harry**	**201010**
333	**Stephan**	**02228**

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444	**Lan**	**60007**
555	**Katharine**	**06389**
666	**John**	**462007**

Superkeyinthetableabove:

{EMP_ID},{EMP_ID,EMP_NAME},{EMP_ID,EMP_NAME,EMP_ZIP}.....soon.

Candidatekey:{EMP_ID}

Non-primeattributes:Inthegiventable,allattributesexceptEMP_IDarenon- prime.

Here, EMP_STATE & EMP_CITY dependent on EMP_ZIP and EMP_ZIP dependent on EMP_ID.The non-prime attributes (EMP_STATE, EMP_CITY) transitively dependent on super key (EMP_ID). It violates the rule of third normal form.

That'swhyweneedtomovetheEMP_CITYandEMP_STATEtothenew <EMPLOYEE_ZIP>table,withEMP_ZIPasaPrimarykey.

EMPLOYEE Table: -

EMP_ID	EMP_NAME	EMP_ZIP
222	**Harry**	**201010**
333	**Stephan**	**02228**
444	**Lan**	**60007**
555	**Katharine**	**06389**

666	John	462007

EMPLOYEE_ZIPTable:

EMP_ZIP	EMP_STATE	EMP_CITY
201010	UP	Noida
02228	US	Boston
60007	US	Chicago
06389	UK	Norwich
462007	MP	Bhopal

Itisanadvanceversionof3NFthat'sitisalsoreferredas 3.5NF.

AtablecomplieswithBCNFifitisin3NFandforeveryfunctionaldependencyX->Y,Xshould bethesuperkeyof the table.

FunctionalDependency

Functional Dependency (FD) determines the relation of one attribute to anotherattribute in a database management system (DBMS) system.

Functionaldependencyhelpsyoutomaintainthequalityofdatainthedatabase.

A functional dependency is denoted by an arrow →. The functional dependencyof X on Y is represented by X → Y.

Boyce-CoddNormalForms, ThirdNormalForm

Example:Supposethereisacompanywhereinemployeesworkin morethan one department. They store the data like this:

emp_id	emp_nationality	emp_dept	dept_type	dept_no_of_emp
1001	Austrian	Production and planning	D001	200
1001	Austrian	stores	D001	250
1002	American	designand technical support	D134	100
1002	American	Purchasing department	D134	600

Functionaldependencies inthetableabove:
emp_id->emp_nationality

emp_dept->{dept_type,dept_no_of_emp}.

Candidate key: {emp_id, emp_dept}

ThetableisnotinBCNFasneitheremp_id noremp_deptaloneare keys.

TomakethetablecomplywithBCNFwecan breakthetableinthreetableslikethis:

emp_nationality:

emp_id	emp_nationality
1001	Austrian
1002	American

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emp_depttable:

emp_dept	dept_type	dept_no_of_emp
Productionandplanning	D001	200
Stores	D001	250
designandtechnicalsupport	D134	100
Purchasingdepartment	D134	600

emp_dept_mappingtable:

emp_id	emp_dept
1001	Productionandplanning
1001	Stores
1002	designandtechnicalsupport
1002	Purchasingdepartment

Functionaldependencies:
emp_id->emp_nationality

emp_dept->{dept_type,dept_no_of_emp}

Candidate keys:
Forfirsttable:emp_id

Forsecondtable:emp_dept

Forthirdtable:{emp_id,emp_dept}

ThisisnowinBCNFasinboththefunctionaldependenciesleftsidepartisa key.

References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. FundamentalsofDatabaseSystems"byElmsari,Navathe,5thEdition,Pearson Education (2008).

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UNIT-3(FourthandFifthNormalForm) LECTURE NO: -25

Fourthnormalform(4NF)

- A relation will be in 4NF if it is in Boyce Codd normal form and hasno multi-valued dependency.
- For a dependency A → B, if for a single value of A, multiple values of Bexist, then the relation will be a multi-valued dependency.

MultivaluedDependency

Multivalueddependency occurs when two attributes in a table areindependentof each other but, both depend on a third attribute.

Amultivalueddependency consists ofat least two attributes that aredependent ona third attribute that's why it always requires at least three attributes.

Example: Suppose there is a bike manufacturer company which produces two colors (white and black) of each model every year.

BIKE_MODEL	MANUF_YEAR	COLOR
M2011	2008	White
M2001	2008	Black
M3001	2013	White
M3001	2013	Black
M4006	2017	White
M4006	2017	Black

- Here columns COLOR and MANUF_YEAR are dependent on BIKE_MODEL and independent of each other.
- In thiscase, these two columnscanbecalled asmultivalueddependenton BIKE_MODEL. The representation of these dependencies is shown below:
- BIKE_MODEL→→MANUF_YEAR
- BIKE_MODEL→→COLOR

- This can be read as "BIKE_MODEL multidetermined MANUF_YEAR" and "BIKE_MODEL multidetermined COLOR".

Example:student

STU_ID	COURSE	HOBBY
21	Computer	Dancing
21	Math	Singing
34	Chemistry	Dancing
74	Biology	Cricket
59	Physics	Hockey

- The given STUDENT table is in 3NF, but the COURSE and HOBBY are two independent entity. Hence, there is no relationship between COURSE and HOBBY.
- IntheSTUDENTrelation,astudentwithSTU_ID, 21 containstwo courses,Computer and Mathand two hobbies,Dancingand Singing. So there is a Multi-valued dependency on STU_ID, which leads to unnecessary repetition of data.

  Sotomaketheabovetableinto4NF,wecandecomposeitintotwotables:

STUDENT_COURSE

STU_ID	COURSE
21	Computer
21	Math
34	Chemistry
74	Biology
59	Physics

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STUDENT_HOBBY

STU_ID	HOBBY
21	Dancing
21	Singing
34	Dancing
74	Cricket
59	Hockey

JoinDependency

Joint dependency – Join decomposition is a further generalization of Multivalued dependencies.

If the join of R1 and R2 over C is equal to relation R then we can say that a join dependency (JD) exists,where R1andR2are the decompositionR1(A,B,C) andR2(C, D) of a given relations R (A, B, C, D).

Alternatively,R1andR2area losslessdecompositionofR

Example–TableR1

COMPANY	PRODUCT
C1	pen drive
C1	mic
C2	speaker
C2	speaker

TableR2

AGENT	COMPANY
Aman	C1
Aman	C2
Mohan	C1

Agent->->CompanyCompany->->Product

TableR3

AGENT	PRODUCT
Aman	pendrive
Aman	mic
Aman	speaker
Mohan	speaker

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Agent->->Product

Table–R1⋈R2⋈R3

COMPANY	PRODUCT	AGENT
C1	pen drive	Aman
C1	mic	Aman
C2	speaker	speaker
C1	speaker	Aman

FifthNormalForm/ProjectedNormalForm(5NF):-

- ArelationRisin5NF ifandonly ifeveryjoindependencyinR isimpliedby the candidate keys of R.
- A relation decomposed into two relations must have loss-less join Property, which ensures that no spurious or extra tuples are generated, when relations are reunited through a natural join.
- **Properties** – A relation R is in 5NF if and only if it satisfies followingconditions:
- Rshouldbealreadyin4NF.
- Itcannotbefurthernonlossdecomposed(joindependency).

Example – Consider the above schema, with a case as “if a company makes a product and an agent is an agent for that company, then he always sells that productforthe company”.Underthesecircumstances,theACPtable isshown as:

Table–ACP

AGENT	COMPANY	PRODUCT
A1	PQR	Nut
A1	PQR	Bolt
A1	XYZ	Nut
A1	XYZ	Bolt
A2	PQR	Nut

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The relation ACP is again decompose into3 relations. Now,the naturalJoin of all the three relations will be shown as:

Table–R1

AGENT	COMPANY
A1	PQR
A1	XYZ
A2	PQR

Table- R2

AGENT	PRODUCT
A1	Nut
A1	Bolt
A2	Nut

Table–R3

COMPANY	PRODUCT
PQR	Nut
PQR	Bolt
XYZ	Nut
XYZ	Bolt

- ResultofNaturalJoinofR1andR3over'Company'andthen NaturalJoinof R13 and R2 over 'Agent 'and 'Product' will be table ACP.
- Hence,inthisexample,alltheredundanciesareeliminated,andthe decomposition of ACP is a lossless join decomposition.
- Therefore, the relation is in 5NF as it does not violate the property of lossless join.

References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. FundamentalsofDatabaseSystems"byElmsari,Navathe,5thEdition,Pearson Education (2008).

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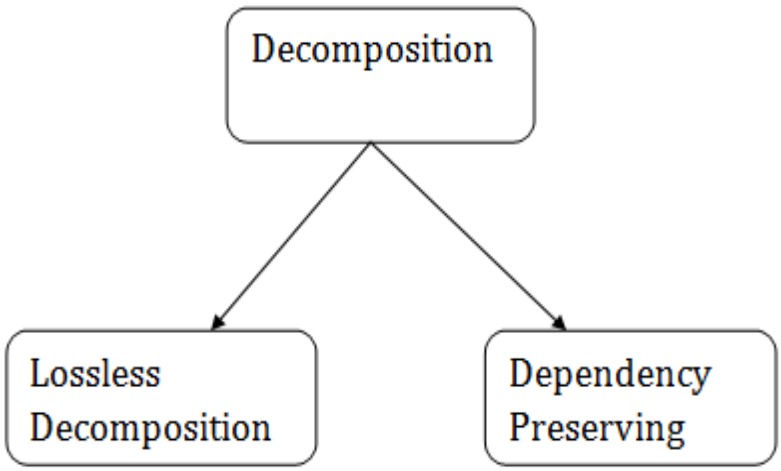
UNIT-3(DecompositionintoBCNFand3-NF)
LECTURE NO: -26

RelationalDecomposition

When a relation in the relational model is not in appropriate normal form then the decomposition of a relation is required.

In a database, it breaks the table into multiple tables. If the relation has no proper decomposition, then it may lead to problems like loss of information. Decomposition is used to eliminate some of the problems of bad design like anomalies, inconsistencies, and redundancy**.**

TypesofDecomposition



LosslessDecomposition

- Iftheinformationisnotlostfromtherelationthatisdecomposed,thenthe decomposition will be lossless.
- The lossless decomposition guarantees that the join of relations will result in the same relation as it was decomposed.
- Therelationissaidtobelosslessdecompositionifnaturaljoinsofallthe decomposition give the original relation**.**

EMPLOYEE_DEPARTMENTtable:

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY	DEPT_ID	DEPT_NAME
22	Denim	28	Mumbai	827	Sales
33	Alina	25	Delhi	438	Marketing
46	Stephan	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production
60	Jack	40	Noida	678	Testing

TheaboverelationisdecomposedintotworelationsEMPLOYEEandDEPARTMENT

EMPLOYEEtable:

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY
22	Denim	28	Mumbai
33	Alina	25	Delhi
46	Stephan	30	Bangalore
52	Katherine	36	Mumbai
60	Jack	40	Noida

DEPARTMENTtable

DEPT_ID	EMP_ID	DEPT_NAME
827	22	Sales
438	33	Marketing
869	46	Finance
575	52	Production
678	60	Testing

Now, when these two relations are joined on the common column "EMP_ID", then the resultant relation will look like:

Employee⋈Department

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY	DEPT_ID	DEPT_NAME
22	Denim	28	Mumbai	827	Sales
33	Alina	25	Delhi	438	Marketing
46	Stephan	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production

60	Jack	40	Noida	678	Testing

Hence,thedecompositionisLosslessjoindecomposition.

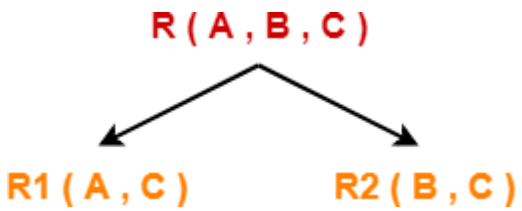
DependencyPreserving

- Itisanimportantconstraint ofthe database.
- In the dependency preservation, at least one decomposed table must satisfy every dependency.
- IfarelationR is decomposed intorelationR1andR2,thenthe dependencies ofR either must be a part of R1 or R2 or must be derivable from the combination of functional dependencies of R1 and R2.
- For example, suppose there is a relation R (A, B, C, D) with functionaldependency set (A->BC). The relational R is decomposed into R1(ABC) andR2(AD) which is dependency preserving because FD A->BC is a part of relation R1(ABC).

A	B	C
1	2	1
2	5	3
3	3	3

R(A,B,C)

- ConsiderthisrelationisdecomposedintotwosubrelationsasR_1(A,C)and R_2(B, C)-



- Thetwosubrelationsare-

R_1(A,C)

A	C
1	1
2	3
3	3

R_2(B,C)

B	C
2	1
5	3
3	3

- Now,letuscheck whetherthisdecompositionis lossyornot.
- Forlossydecomposition,wemusthave-

  $R_1 \bowtie R_2 \supset R$
- Now,ifweperformthenaturaljoin(⋈) ofthesubrelationsR_1andR_2weget-

A	B	C
1	2	1
2	5	3
2	3	3
3	5	3
3	3	3

References:-

1. DatabaseSystemConcepts6thEditionAbrahamSilberschatz,HenryF.Korth.
2. FundamentalsofDatabaseSystems”byElmsari, Navathe,5thEdition, Pearson Education (2008).